

Scared Together: Heart Rate Synchrony and Social Closeness in a High-Intensity Horror Setting

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Humans frequently seek fear-inducing entertainment in social settings, yet the influence of social dynamics on physiological and emotional responses to shared fear remains underexplored. In this field study, we examined physiological synchrony and subjective responses to a high-intensity horror setting at a haunted house attraction. Groups of visitors ($N = 347$, aged 18–63 years) wore heart rate (HR) monitors and reported their emotional experiences ($N = 347$ for self-reports; $N = 254$ for HR analyses). HR synchrony emerged during the haunted house experience across timescales from short (25 s) to long (~55 min), and greater HR synchrony was associated with increased subjective arousal. Additionally, socially close dyads exhibited higher HR synchrony compared with less close pairs. These findings indicate that physiological synchrony is linked to subjective arousal during shared frightening experiences and is more pronounced among socially close individuals.

Keywords: fear, heart rate synchrony, arousal, valence, horror

Despite their apparent paradoxical nature, frightening entertainment and recreational forms of fear are historically pervasive and widespread social phenomena. The peculiar appeal of entertainment that evokes negative emotions such as fear and anxiety are often referred to as the “paradox of horror” (Carroll, 1990) and has been the subject of much philosophical speculation as well as scientific research (for a review, see Clasen, 2017, 2021; Martin, 2019). Yet, despite the inherently social nature of enjoying

frightening entertainment, little is known about the effects of social dynamics on individual experience and physiology, including how social relationships might modulate such factors.

Many accounts of recreational fear revolve around the idea that frightening entertainment serves as a risk-free and low-cost opportunity for exploring negative emotions like fear and anxiety (Andersen et al., 2020; Clasen et al., 2020; Grodal, 2009; Menninghaus et al., 2017; Miller et al., 2023; Morin et al., 2019). While the primary

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methodology, writing—original draft, and writing—review and editing. Pedro A. M. Mediano played a supporting role in methodology and writing—original draft and an equal role in formal analysis, visualization, and writing—review and editing. Henry Price played a supporting role in data curation and writing—review and editing and an equal role in formal analysis. Ivana Konvalinka played a supporting role in conceptualization and methodology and an equal role in formal analysis, writing—original draft, and writing—review and editing. Mathias Clasen played a lead role in writing—original draft and writing—review and editing, a supporting role in formal analysis, and an equal role in conceptualization, data curation, funding acquisition, methodology, and project administration.

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attractor of recreational horror has traditionally been identified as the physiological arousal and experience of suspense and fear (Martin, 2019), too high relative levels of arousal and fear in frightening entertainment have been associated with overall negative experiences (Clasen, 2021; Harrison & Cantor, 1999). Instead, recent evidence suggests that relatively moderate experiences of fear and physiological arousal are more likely to induce pleasure (Andersen et al., 2020), making engagement with horror akin to other forms of playful behavior (Andersen et al., 2023). The appetite for frightening entertainment seems to increase with age and peaks in the late teenage years (Hoffner & Levine, 2005); and has been found to correlate with traits sensation-seeking and openness to experience (Clasen et al., 2020). Recent studies have also identified different motivational profiles of horror fans, for example, so-called adrenaline junkies and white-knucklers, who respectively seek out extreme arousal or challenge themselves to cope with fear and ultimately learn from it, for instance by gaining insight into their own emotional and reactive patterns (Clasen et al., 2019; Robinson et al., 2014; Scrivner et al., 2023).

Engagement with frightening entertainment and recreational fear, however, is inherently a social phenomenon. People tend to watch horror movies together, they travel to haunted attractions in groups, they often play horror video games together and even horror fiction has historically been shared collectively in the form of being narrated aloud (Clasen et al., 2019, 2020; Michaelson, 2010). Even so, the effects of social dynamics on subjective experience in the context of recreational horror remain largely unexplored. On the one hand, group support might enable individuals to endure frightening or threatening stimuli, particularly in the presence of close others, as suggested by social baseline theory (Beckes & Sbarra, 2022). On the other hand, emotional contagion during face-to-face interactions could act as a catalyst, amplifying the intensity of the experience (Clasen et al., 2019; Espinosa et al., 2024; Hornstein & Eisenberger, 2017).

Shared experiences even in the absence of social interaction have also been shown to intensify emotions, amplify subjective experiences, and even affect neural processing, amplifying the neural processing of visual stimuli (e.g., Boothby et al., 2014; Chung et al., 2024; Lomoriello et al., 2022; Shteynberg et al., 2014). Additionally, research suggests that sharing an experience with others facilitates social learning (Shteynberg & Apfelbaum, 2013) and that this effect is particularly strong in people who are socially proximate (Boothby et al., 2016). It has been suggested that this effect might occur due to individuals devoting greater cognitive resources to features of the environment that are believed to be coattended by a socially close other (Shteynberg, 2015). However, there may be additional underlying mechanisms, as some studies have found no amplification effects on affect intensity, attention, or memory due to shared experiences and shared attention (Jolly et al., 2019; Mairon et al., 2020).

Recent research also suggests that physiological responses to shared events may be modulated by social affinity. One study found that an increasing ratio of friends-to-strangers predicted higher levels of arousal in an immersive horror setting (Tashjian et al., 2022). Another study on participants and spectators during a high arousal firewalking ritual found that firewalkers and affiliated spectators synchronized their heart rhythms together, even in the absence of synchronized behavior, while the same did not happen between firewalkers and unaffiliated spectators (Konvalinka et al., 2011). Interpersonal physiological synchrony has also been reported

between other affiliated pairs, such as mother–infant dyads (Feldman et al., 2011; McFarland et al., 2020), romantic couples (P. Goldstein et al., 2017; Helm et al., 2012), and attracted blind dates (Prochazkova et al., 2022), and during various other shared experiences (for a review, see Palumbo et al., 2017) such as choir singing (Müller & Lindenberg, 2011), playful joint actions (Fusaroli et al., 2016; Noy et al., 2015), and coattended movies (Golland et al., 2015). While the function of interpersonal physiological synchrony is unknown, it has been proposed to have a role in the formation or maintenance of social bonds (Feldman et al., 2011; Gordon et al., 2020) and the sharing of socially arousing experiences (Golland et al., 2015; Konvalinka et al., 2011). Some support for this comes from previous work on physiological linkage, showing that during periods of shared positive emotion, married couples exhibit higher physiological linkage, measured using Pearson correlations (Chen et al., 2021). Interestingly, this physiological linkage was associated with higher quality interactions and relationships.

To address the knowledge gap about the potential role of social dynamics during shared recreational fear experiences, we took advantage of the high-intensity, immersive, and social recreational horror setting of a commercial haunted house to investigate the relationship between physiological synchrony, arousal, and subjective experience, as well as the effects of social closeness on physiological synchrony. Visitors of varying degrees of social closeness ($N = 347$, Age = 18–63 years) entered in groups (of 3–5) and were equipped with heart rate (HR) monitors and asked to report on their subjective experience of arousal and fear. We chose to measure HRs because they provide a reliable and practical measure of physiological responses in real-world settings, due to the ease of measurement and feasible synchronization between devices. In addition, previous research has established HR modulation as a physiological correlate of self-reported emotions (Cacioppo et al., 2000; Critchley et al., 2005; Davis, 1992; Myrtek & Brügger, 1996), and HR synchrony as a marker of synchronized arousal (Konvalinka et al., 2011) and engagement (Pérez et al., 2021), particularly between socially close partners (Feldman et al., 2011; P. Goldstein et al., 2017; Konvalinka et al., 2011).

We tested three main hypotheses informed by the literature. Building on prior research showing that shared attention to emotional stimuli synchronizes physiological signals between coattending participants (Golland et al., 2015) and intensifies subjective experiences and emotional responses (Boothby et al., 2014; Shteynberg et al., 2014), we (a) expected interpersonal physiological synchrony to emerge during the haunted house experience and (b) for physiological synchrony to be associated with subjective arousal and valence. Finally, based on findings that social bonds and affiliation facilitate physiological synchrony between individuals (Konvalinka et al., 2011; Prochazkova et al., 2022), we (c) predicted that socially close pairs will exhibit higher physiological synchrony compared to pairs who were not close or were strangers.

Method

Ethics and Informed Consent

The methodology in the present study was carried out in accordance with all relevant guidelines and regulations issued by the National Committee on Health Research Ethics, Copenhagen,

Denmark. Informed consent was obtained from all participants prior to their participation.

Participants

Visitors ($N = 347$) signed up for the study on site (192 females, 152 males, three did not respond; $M_{\text{age}} = 29.6$, age range = 18–63). The sample size was determined by the number of guests who were willing to participate in the study, which was conducted over three weekends, totaling six evenings. Participants entered in groups. All participants reported knowing one or more people from the group they went through the attraction with (amount of people known: $M = 2.42$, $SD = 1.12$). Two hundred ninety-eight participants reported being emotionally close to one or more people from the group. People generally reported that there was a high likelihood that they would complete the attraction without dropping out (“What is the likelihood that you will complete the entire attraction today?” Scale from 0% to 100%: $M = 87.23\%$, $SD = 20.70\%$).

Procedure

The data were collected at a Danish commercial haunted attraction called “Dystopia Haunted House,” in which paying visitors went through 50 thematically connected rooms to experience a frightening, interactive, and live-action theatrical horror production (for an example, see Figure 1). The attraction roughly had a 5% dropout rate, where guests chose to abort the attraction midway due to perceiving the attraction as too scary. Guests went through in groups consisting of a combination of friends and strangers. Each participant was inside the haunted attraction for roughly 55 min ($M = 55.46$ min, $SD = 5.42$ min).

Visitors were recruited for the study on arrival at the site and were entered into a lottery for 500 Danish kroner (~U.S.\$80) gift certificates for participating. Participants signed a consent form, were equipped with HR monitors (see the HR Monitors subsection), and

completed a short, structured questionnaire (see the Structured Questionnaires section). Participants then completed the haunted house in groups like regular visitors (87 groups, participants per group: $M = 3.92$ $SD = 0.85$).¹ When participants exited the haunted house, they were asked to complete a Structured Questionnaire (see the Structured Questionnaires section) and return their HR monitors.

Measures and Equipment

Structured Questionnaires

Participants completed two structured questionnaires. The first questionnaire, which was administered prior to participants entering the attraction, contained questions about (a) age, (b) gender, (c) which other group members were *known* to the participant, (d) which other group members were *socially close* to the participant (e.g., a best friend, significant other), and (e) the participant’s own estimated likelihood of completing the attraction. The second questionnaire, which was administered to participants when they exited the attraction, asked participants to indicate the valence (“How did you feel while you were inside Dystopia?”) and arousal (“What was your bodily state while inside Dystopia?”) of their overall experience using a 10-point Likert adaption of the Visual Analogue Self-Assessment Manikin² (Bradley & Lang, 1994; Lang, 1980; see Appendix Figure A1). In the present study, the scale was reversed (negative-to-positive valence and low-to-high arousal; see Appendix). Each scale was also supplied with 10 circles underneath in which participants could mark their state with an “X”. Each scale was also supplied with labels at each endpoint. The endpoints of the scale for arousal were labeled “extreme calm—extreme arousal” and for valence labeled “extremely unpleasant—extremely pleasant”. All questions and labels were in Danish.

HR Monitors

The HR of participants was measured continuously throughout the haunted attraction using Firstbeat Bodyguard 2 HR monitors (Firstbeat, Jyväskylä, Finland). These compact, lightweight devices were fitted by research assistants directly on the participants’ skin using two chest electrodes before they entered the attraction. The Bodyguard 2 devices rely on electrical signals (electrocardiogram) to detect the precise timing of each heartbeat with a sampling rate of 1,000 Hz. The device calculates beats per minute (BPM) by analyzing intervals between heartbeats (R–R intervals) and outputs HR at a frequency of 1 Hz, providing an updated BPM value every second. All devices were synchronized on site using Firstbeat software, prior to participants queuing for the attraction.

HR has been used by emotion researchers in an effort to investigate physiological correlates of self-reported emotions, including fear (Cacioppo et al., 2000; Critchley et al., 2005; Davis, 1992; Myrtek & Brügger, 1996). In this literature, an increase in BPM is the most frequently reported and reliable physiological indicator of self-reported fear (Kreibig, 2010; Kreibig et al., 2007). In field studies with high ecological validity, such as a haunted



Note. The picture is from an earlier year than the present study’s data collection. Printed with permission from all people in the photograph. Photo courtesy of Dystopia Entertainment. See the online article for the color version of this figure.

¹ Note that in a few instances, groups consisted of guests who participated in the study and one or two guests who did not. Group size numbers here reflect only participants.

² In cases where participants indicated two adjacent values on either valence or arousal, the value closest to 0 was coded.

attraction, BPM calculated once per second provides a practical and reliable measure of physiological arousal. This approach captures HR trends with sufficient resolution to detect meaningful patterns of physiological fluctuations while balancing the need for manageable data interpretation in a dynamic, unpredictable environment. Although noise and artifacts may still influence short-term data, this method ensures an effective compromise between fine detail and practical usability for studying broad arousal patterns.

Preprocessing of HR Data and Exclusion of Participants

Participants were equipped with HR monitors roughly 5–15 min before they queued to enter the actual attraction. Participants used this time to fill out questionnaires. Similarly, participants often continued to wear the HR monitor while they filled out the questionnaire after exiting the haunt. To ensure that the HR data represented only the time participants had spent inside the haunted attraction, research assistants at the entrance and exit of the attraction noted the exact time that participants entered and exited the attraction. These timestamps were used to trim the time series data in order to ensure that the HR data that were analyzed only represented time spent inside the haunted attraction.

All 347 participants were included in the analyses of self-reported measures. However, 22 participants were excluded from HR analyses due to missing or corrupted HR recordings, or instances where HR monitors were mixed up during the field experiment. Artifact detection and correction was used using firstbeat software (Sami et al., 2004). An additional 71 participants were excluded following a visual inspection of HR signal quality. This method was chosen because the artifact detection algorithm provided by the Firstbeat software rendered a percentage of artifacts based on the entire time series, including periods before participants entered the haunted attraction and after they exited, which were not relevant to the study. Visual inspection of the HR plots revealed that artifacts were often concentrated in these nonrelevant periods. Therefore, trimming the time series to include only the time spent inside the haunted attraction was crucial, but the artifact percentages reported by the software remained reflective of the entire time series. Consequently, after exclusions, the artifact percentage for the full time series decreased ($M = 15.8\%$, $SD = 8.9\%$) compared with the initial values before exclusions ($M = 26.6\%$, $SD = 24.3\%$). Of the 71 participants excluded, four were also missing timestamps for attraction entry, leaving a final sample of 254 participants suitable for HR analysis. Full artifact percentage data for untrimmed time series are available in the open data file (see Data Availability).

Data Analysis

The HR time series were aligned and cropped according to timestamps that marked the beginning and end of the experience. The mean HR and HR variance were computed directly from these time series. The degree of HR synchrony between participants belonging to the same group was calculated using the Pearson correlation coefficient between the corresponding HR time series, previously defined as intersubject correlation of HRs (Pérez et al., 2021). To minimize the effect of outliers, the Gaussian copula method (Ince et al., 2017) was applied to each of the two time series in order to transform the histogram of each into a Gaussian distribution before calculating their correlation. HR variance, as used

in this study, refers to the statistical variability of BPM values recorded during the haunted house experience. While “heart rate variability” typically describes variability in R–R intervals and requires clean data with minimal artifacts as well as proper artifact correction methods (as artifact correction methods can distort HR variability [Laborde et al., 2017]), we opted to use BPM-based variance (squared standard deviation) to align with the type of data collected and the study objectives.

In order to test our first hypothesis, whether HR synchrony emerged during the haunted house experience and whether this could be robustly captured using our synchrony measure, that is, intersubject correlation of HR, we contrasted time-aligned dyadic measures of synchrony with surrogate copies of the data, where the alignment was broken via circular temporal shifts. In other words, a Pearson correlation was computed between each participant’s HR time series and another group member’s time series that was circularly shifted by a random amount within the segments (i.e., the length of the entire haunted house experience), repeating this shuffle 10,000 times, following previous research (Pérez et al., 2021). Each dyad’s time-locked correlation was then compared with this distribution of correlation values for the circularly shifted HR time series, obtaining p values by determining how many circularly shifted values were above the real (time-locked data) values, corrected for multiple comparisons with false discovery rate of 0.01. The circular shuffle was thus used as a control to determine whether synchrony was present.

To complement this analysis, we also compared the HR synchrony values across segments of different lengths. Concretely, for every dyad, we calculated the average Pearson correlation over segments of specific sizes and then contrasted these against averages of correlations obtained with data of nonaligned segments, obtaining p values by determining how many correlation values from non-aligned segments were below the real (time-locked data) values, corrected for multiple comparisons with false discovery rate of 0.01. This additional analysis allowed us to investigate whether synchrony emerged across short and long windows, allowing us to gain more insight into the underlying mechanisms, for example, whether it may be driven by salient events and jump scares, or by more time-grained processes such as attentional ones.

In order to test our second hypothesis, we constructed the following ordinal regression models to test associations between HR synchrony and self-reported measures of arousal and valence. In these models, HR synchrony is estimated per subject by averaging the intersubject correlation values across all dyads in which the subject participated.³ For each dyad, the correlation was calculated across the entire duration of the HR recording inside the haunted attraction. *Model 1a*: Self-reported arousal as dependent variable, predicted by the average synchronization between each participant and all other members of the group. *Model 1b*: Self-reported valence as dependent variable, predicted by the average synchronization between each participant and all other members of the group. Models 1a and 1b used age and gender as controlling variables, and random intercepts to model the nesting of participants within groups. In addition, in order to rule out properties of the HR signals as drivers of the association between HR synchrony and subjective arousal/valence, we also added mean HR and HR variance as

³ In other words, each participant’s synchrony score was derived from their individual pairwise synchrony with every other group member.

controlling variables in a follow-up comparison. These models without the covariates are reported in [Appendix](#).

In order to test our third hypothesis, we constructed the following linear mixed effect model to test the effect of social closeness on HR synchrony, with HR synchrony measured on a dyadic level across the entire duration of the HR recording inside the haunted attraction. *Model 2a*: HR synchrony as dependent variable, with self-reported social closeness (i.e., whether at least one member of the dyad reported to be socially close to the other member of the dyad or not) as a fixed effect, with group number and participant number as random effects. In order to check whether this association could also be explained by mere familiarity, we constructed the following mixed effect model to test the effect of familiarity on HR synchrony: *Model 2b*: HR synchrony as dependent variable, with familiarity (i.e., if the dyad knew each other or not) as a fixed effect, with group number and participant number as random effects. These models without the covariates are reported in [Appendix](#).

In order to control for perceived similarity driving interpersonal synchrony, both models accounted for whether the dyads were of the same gender and whether the age difference was less than 5 years, helping to ensure participants were in similar life stages. To address the nested structure of the HR data, the models include random intercepts to account for the effect of the identity of each of the members of the dyad, and also random intercepts for the group to which they belong.

Finally, as a proof of concept, we conducted three secondary analyses. First, we hypothesized that arousal would be negatively correlated with valence, given the study's context within a frightening haunted attraction. To test this hypothesis, we conducted a Spearman correlation analysis to examine the association between self-reported arousal and self-reported valence. Continuing this proof-of-concept analysis, we further explored the relationship between physiological responses and self-reported emotional experiences. Specifically, we examined whether mean HR and HR variance could predict self-reported arousal and valence. In order to test this, we constructed the following ordinal regression models. *Model 3a*: self-reported arousal as the dependent variable, with mean HR and HR variance as predictors. This analysis aimed to determine whether physiological indicators of heightened activation corresponded to subjective experiences of arousal. *Model 3b*: self-reported valence as the dependent variable, again using mean HR and HR variance as predictors, to test whether variations in HR metrics were associated with the emotional valence participants reported (Model 3b). All the signal-processing procedures were carried out using Python (3.9.1), and the statistical analyses using RStudio (V. 1.3.1093). Statistics of ordinal regression models were calculated using the R package MASS (Venables & Ripley, 2002) and Ordinal (Christensen, 2023), and the statistics of the linear mixed models were calculated using the R package *lmerTest* (Kuznetsova et al., 2015). All numerical variables were normalized so their regression coefficients correspond to standardized betas. Supplementary and variant regression models, including both covariate-adjusted and unadjusted versions of the main analyses, are presented in [Appendix Tables A1–A6](#).

Transparency and Openness

We report how we determined our sample size, all data exclusions, and all measures in the study. The data generated and analyzed during the present study are available at OSF (<https://osf.io/hj4k6/>).

[view_only=94749b0676d2408b972e49b846d6afdd](#)). The study was not preregistered.

Results

Emergence of Interpersonal Physiological Synchrony During the Haunted House Experience

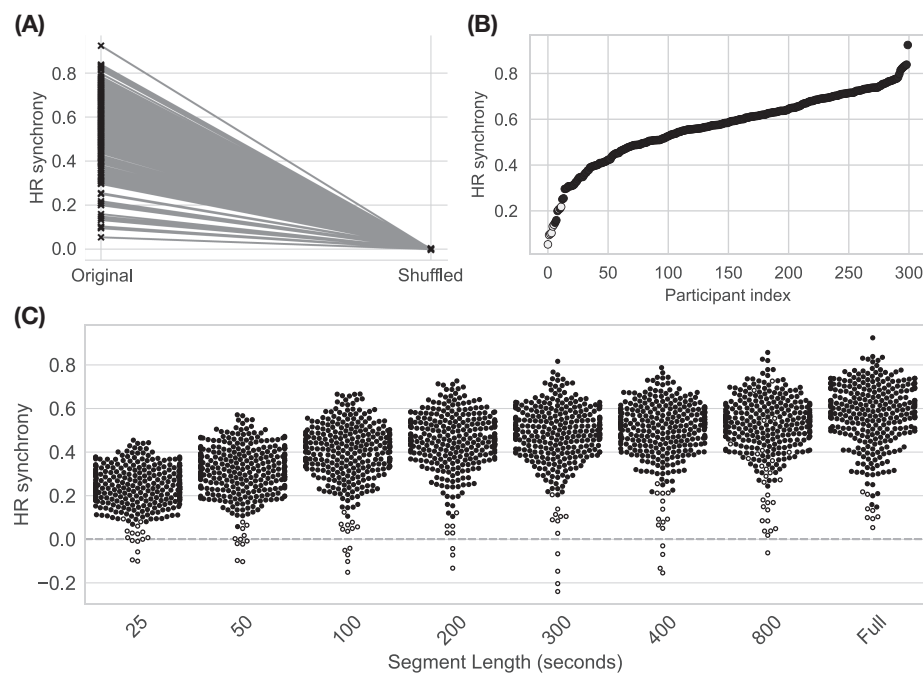
In order to establish whether interpersonal physiological synchrony emerged, we contrasted Pearson correlation values between dyadic time-locked HR time series with circularly shuffled time series (see the Method section). HR synchrony was significant across 292/300 dyads (97.3%) after false discovery rate correction, see [Figure 2A](#) and [2B](#), as determined using circular shuffle statistics. In addition, [Figure 2C](#) shows the effect of segment lengths of the time-series on HR synchrony, with black dots indicating statistically significant values using circular shuffle statistics. While HR synchrony somewhat increases with increasing segment lengths, saturating around 200 s, relatively high levels of synchrony are already present at segment lengths of 25 s. This indicates that the synchrony is likely not only driven by salient events (e.g., jump scares) but also by processes that occur over shorter time-scales, for example, interactions, shared attention, and so forth.

Association Between HR Synchrony and Subjective Arousal and Valence

In order to investigate whether people who were more physiologically synchronous with other group members experienced higher arousal, we analyzed the relationship between HR synchrony and self-reported arousal, controlling for age and gender. For this, we calculated the degree of synchrony per participant by averaging the amount of synchrony across all the dyads the participant took part in (see the Method section). Results revealed a statistically significant association between HR synchrony and self-reported arousal (Model 1a: $\beta = 0.44$, $SE = .13$, $p < .001$; see [Figure 3](#)), with participants who were more synchronous with their group members reporting higher arousal. This model controlled for the effects of gender and age, with males reporting lower arousal (Model 1a: $\beta = -1.09$, $SE = 0.26$, $p < .001$), and age being negatively associated with self-reported arousal (Model 1a: $\beta = -0.33$, $SE = 0.13$, $p = .013$). The association between synchrony and subjective arousal was reduced to a statistical trend when controlling for the mean HR and HR variance of participants (see [Appendix Table A1](#)).

In order to investigate whether people who were more physiologically synchronous with other group members also had more affective (e.g., fearful) experiences, we also tested the association between HR synchrony and reported valence. There was no statistically significant association between HR synchrony and self-reported valence, although there was a statistical trend (Model 1b: $\beta = -0.26$, $SE = 0.16$, $p = .096$). These models controlled for the effects of gender and age, with males reporting more positive valence compared with females (Model 1b: $\beta = 0.91$, $SE = 0.27$, $p < .001$), and age having no statistically significant effect on self-reported valence (Model 1b: $\beta = 0.15$, $SE = 0.15$, $p = .309$). The statistical trend between HR synchrony and self-reported valence was no longer present when controlling for mean HR and HR variance of participants (see [Appendix Table A2](#)).

Figure 2
Emergence of Interpersonal Physiological Synchrony



Note. (A) HR synchrony measured using Pearson correlation is significantly higher between dyadic time-locked time series than circularly shuffled time series (false discovery rate correction of 0.01). (B) The dyadic HR synchrony data are ordered by their correlation strength. (C) Effect of segment length on HR synchrony. The full HR recording is an average length of 3,327 s ($SD = 325.4$ s). Black dots indicate statistically significant values, while unfilled dots indicate nonsignificant values. HR = heart rate.

Modulation of Interpersonal HR Synchrony by Social Closeness

In order to investigate whether socially close dyads had higher HR synchrony than dyads who were not close, we investigated the effect of social closeness on HR synchrony at a dyadic level. Results revealed that dyads who reported having a close social relationship had higher interpersonal HR synchrony than dyads who were not close (Model 2a: $\beta = 0.25$, $SE = 0.11$, $p = .020$; see Figure 4). In contrast, there was no statistically significant effect of merely knowing each other on HR synchrony in dyads (Model 2b: $\beta = 0.24$, $SE = 0.13$, $p = .066$). The effect between HR synchrony and being socially close was also not explained by people being of similar gender or age, as both models controlled for these factors (see the Method section). There was, however, a separate statistically significant contribution from being the same gender on HR synchrony in dyads (Model 2a: $\beta = 0.20$, $SE = 0.10$, $p = .032$), but no effect of being the same age (Model 2a: $\beta = 0.19$, $SE = 0.12$, $p = .111$).

Self-Report Measures of Subjective Emotional Experience

As expected, there was a statistically significant negative Spearman correlation between self-reported arousal and self-reported valence ($r = -0.40$, $p < .001$). This result suggests that participants who reported higher levels of arousal tended to

experience more negative valence, consistent with the emotional dynamics expected in a haunted attraction setting.

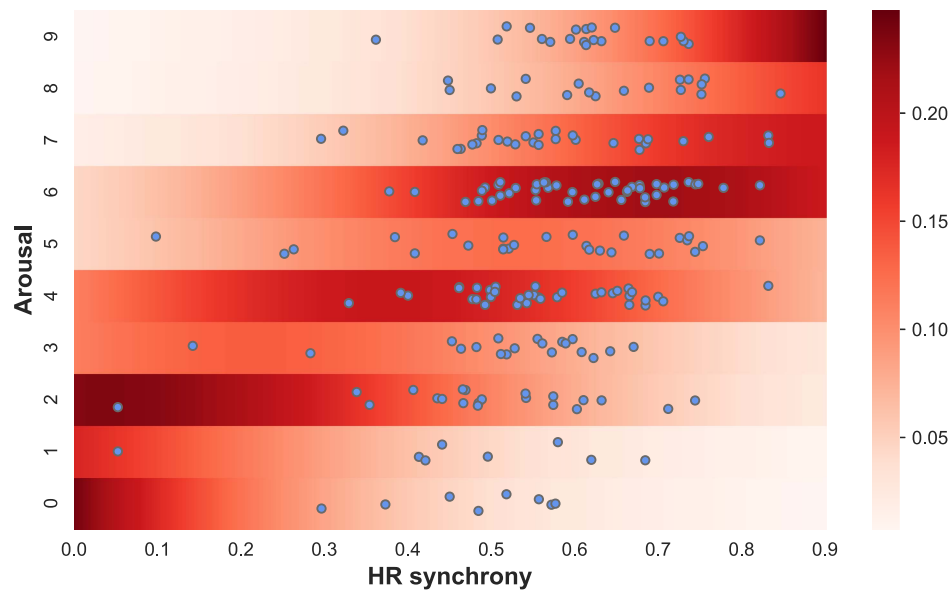
Mean HR, HR Variance, and Subjective Emotional Experience

We found a significant linear relationship between mean HR and self-reported arousal (Model 3a: $\beta = 0.61$, $SE = 0.13$, $p < .001$), and a statistically significant relationship between HR variance and self-reported arousal (Model 3a: $\beta = 0.45$, $SE = 0.12$, $p < .001$; see Figure 5). There was no statistically significant relationship between mean HR and valence (Model 3b: $\beta = -0.22$, $SE = 0.12$, $p = .054$). There was a significant negative association between HR variance and valence (Model 3b: $\beta = -0.40$, $SE = 0.12$, $p < .001$).

Summary of Results

For our three main hypotheses, results revealed (a) that interpersonal HR synchrony emerged during the haunted house experience and was present across both shorter (25 s) and longer (~55 min) time scales. Moreover, (b) increased HR synchrony was associated with increased subjectively experienced arousal. Finally, results also revealed that (c) close social relationships in dyads were associated with increased synchrony in HR.

Figure 3
Association Between HR Synchrony and Arousal



Note. Participants that exhibited a higher degree of HR synchrony with other participants tended to report higher arousal scores. Color represents probability of each arousal value predicted with an ordinal regression model using HR synchrony as predictor. A small amount of jitter is added to the Y axis for visualization purposes. HR = heart rate. See the online article for the color version of this figure.

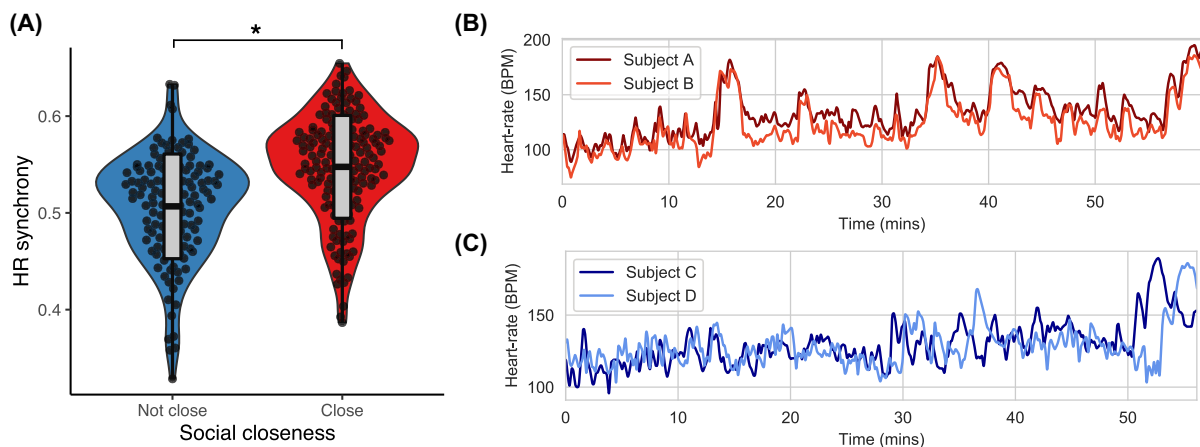
For our secondary hypotheses, our results revealed, as expected, a negative association between self-reported arousal and valence, consistent with the emotional dynamics expected in a haunted attraction (i.e., fear). As expected, results also revealed a significant relationship between mean HR and self-reported arousal as well as between HR variance and self-reported arousal. While we expected to find a statistically significant negative relationship between mean HR and valence, this relationship was marginally nonsignificant.

However, results did reveal a statistically significant negative relationship between HR variance and valence.

Discussion

In this study, we examined the relationship between physiological synchrony, arousal, and subjective experience within a high-intensity, immersive, and social recreational horror setting. Our results

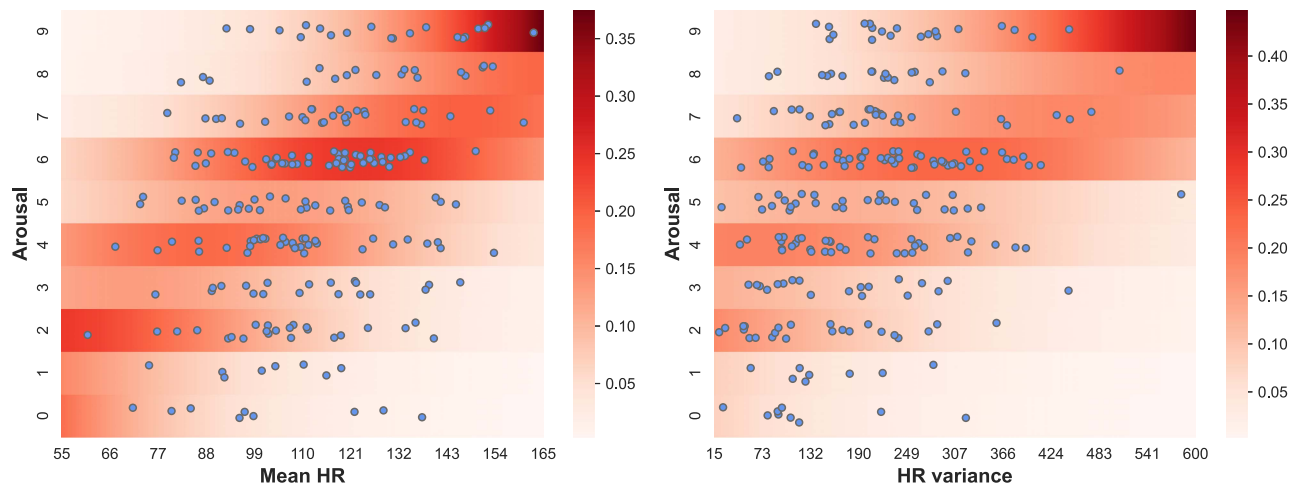
Figure 4
Higher HR Synchrony in Socially Close Dyads



Note. (A) Relationship between reported social closeness and heart rate synchrony. (B) An example of a highly synchronized (and socially close) dyad. (C) An example of a less synchronized (and not socially close) dyad. HR = heart rate; BPM = beats per minute. See the online article for the color version of this figure.

* $p = .020$.

Figure 5
Mean HR and Variance Predict Subjective Arousal



Note. Mean HR (left) and HR variance (right) were found to be positively correlated with self-reported arousal. Color represents probability of each arousal value predicted with an ordinal regression model using mean HR (left) or HR variance (right) as predictors. A small amount of jitter is added to the Y axis for visualization purposes. HR = heart rate. See the online article for the color version of this figure.

demonstrate that interpersonal HR synchrony is detectable as a strong signal in a real-life setting, with correlations observed at both short and long segment lengths, indicating that the synchrony is consistent across the entire time scale and not solely driven by one singular or prominent event (Figure 2C). Moreover, we found that participants who synchronized their HRs with others in their group reported higher arousal. This synchrony was more pronounced among socially close dyads, suggesting that social closeness may facilitate physiological alignment during shared emotional experiences. Although we observed a positive association between interpersonal HR synchrony and self-reported arousal, this relationship disappeared when controlling for mean HR and HR variance. This suggests that the observed synchrony-arousal link may, in part, reflect the influence of overall HR dynamics—where elevated or more variable HRs both enhance synchrony and are independently associated with higher subjective arousal. Overall, these findings highlight the potential role of physiological synchrony in shaping subjective and physiological responses during high-arousal, shared frightening experiences.

Our findings that increased interpersonal HR synchrony was associated with increased subjectively experienced arousal align with previous literature documenting amplification effects of shared experiences (Boothby et al., 2014), as well as literature showing synchronization of physiological signals between interacting individuals (Palumbo et al., 2017), and participants jointly attending to the same emotional stimuli (Golland et al., 2015). In particular, previous studies have shown that sharing experiences with others amplifies subjective experiences (Boothby et al., 2014), intensifies emotional responses (Shteynberg et al., 2014), and amplifies the neural processing of visual input (Lomoriello et al., 2022). Importantly, this amplification is modulated by the psychological distance between people, such that shared experiences are particularly amplified between people who are psychologically proximate (Boothby et al., 2016).

Intriguingly, our results suggesting that close social relationships in dyads are associated with increased synchrony of HRs support

previous evidence that HRs are more synchronized between close friends, partners, and family than between unaffiliated members, in contexts of highly arousing events (Konvalinka et al., 2011). Other studies similarly show increased physiological synchrony between affiliated dyads (Feldman et al., 2011; S. Goldstein et al., 1989, 2017; Helm et al., 2012; Prochazkova et al., 2022). For example, S. Goldstein et al. (1989) found that among toddlers and preschoolers, friends synchronized more than acquaintances during play (greater concordance on baseline HR and cortisol). One key question that remains is how much of this interpersonal synchrony emerges between socially close dyads during shared experiences, and how much is already present even at baseline levels. In other words, although our results suggest that close social relationships in a recreational horror setting facilitate physiological synchrony, it could be that physiological synchrony is already present between socially close participants (at higher levels than between unaffiliated participants) before they enter the haunted house. Alternatively, it could be that socially close dyads stay closer together, hence getting a more time-locked and emotionally aligned haunted house experience.

Another potential explanation for the presence of interpersonal physiological synchrony is through attentional processes. Previous research has shown that people's physiological signals (e.g., HRs, eye movements, neural signals) become correlated when they attend to the same narrative stimuli, both visual and auditory, even in the absence of interaction or a shared social context (Hasson et al., 2004; Madsen & Parra, 2022; Pérez et al., 2021). In particular, Pérez et al. (2021) showed that narrative auditory stimuli synchronized HRs between individuals, measured as intersubject (Pearson) correlation, at short 1-min length segments, driven by common stimulus locking. Importantly, intersubject HR correlation was modulated by attention, such that people's HR signals become more correlated during periods of joint attention and less during periods of distraction (Pérez et al., 2021), suggesting that interpersonal synchrony may be a marker of joint attention and engagement.

In addition, previous work has shown that stimulus-driven intersubject correlation is higher among friends (Parkinson et al., 2018). Parkinson et al. (2018) measured the similarity in neural responses between friends and strangers while watching the same movies and found evidence of neural homophily—that friends had more similar neural responses to the movies compared with strangers, even in the absence of any social context or interaction between them.

There may thus be multiple mechanisms underlying the presence or emergence of interpersonal physiological synchrony, related to common stimulus locking, increased attentional processing when sharing experiences particularly with friends, or more similarity between friends in their physiological or emotional responses to stimuli.

Not surprisingly, our results also reveal positive relationships between mean HR and self-reported arousal as well as HR variance and self-reported arousal. Similarly, our results revealed a nonsurprising negative association between HR variance and self-reported valence. These results indicate an expected correspondence between subjectively felt fear and physiological arousal, with comparable relationships being found across numerous studies (for a review, see Kreibitz et al., 2007). Likewise, our results reveal a negative relationship between age and self-reported arousal, as well as negative relationships between gender and self-reported arousal and valence, with females generally reporting higher arousal and more negative valence, which has also been found in previous studies (Andersen et al., 2020; Clasen et al., 2019).

Our study has several limitations. We had to exclude a relatively high number of participants (93 excluded out of 347) from the HR analyses due to a variety of methodological challenges. First, because of the high-intensity horror setting, participants were often startled and made a variety of sudden movements, including colliding with or gripping other group members. Similarly, participants would sometimes perspire extensively when exiting the attraction. These factors contribute to the likelihood of disruption of the HR signal detection, as attested by the number of participants excluded due to missing or corrupted HR data. Likewise, the dark and boisterous environment may have contributed to a few instances of data loss, such as a few participants mixing up their HR monitors before returning them and a group of participants that were not assigned an entry time by our team.

Another limitation with the study pertains to problems with the self-assessment manikin, which is designed to measure valence and arousal in emotional experience. Several participants informally reported that it was difficult for them to assign either a positive or negative valence to their experience. Many of these participants expressed a desire to indicate that the experience had been both negatively and positively valenced at the same time, reflecting that they had enjoyed being frightened. These reports are compatible with recent findings on mixed emotions in recreational horror (Andersen et al., 2020) and support previous claims that the positive and negative affective bases that underlie valence are best thought of as two separate dimensions (Cacioppo & Berntson, 1994; Kuppens et al., 2013; Larsen & McGraw, 2014). For the context of recreational horror at least, future studies may thus benefit from administering multidimensional measures of valence.

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Appendix

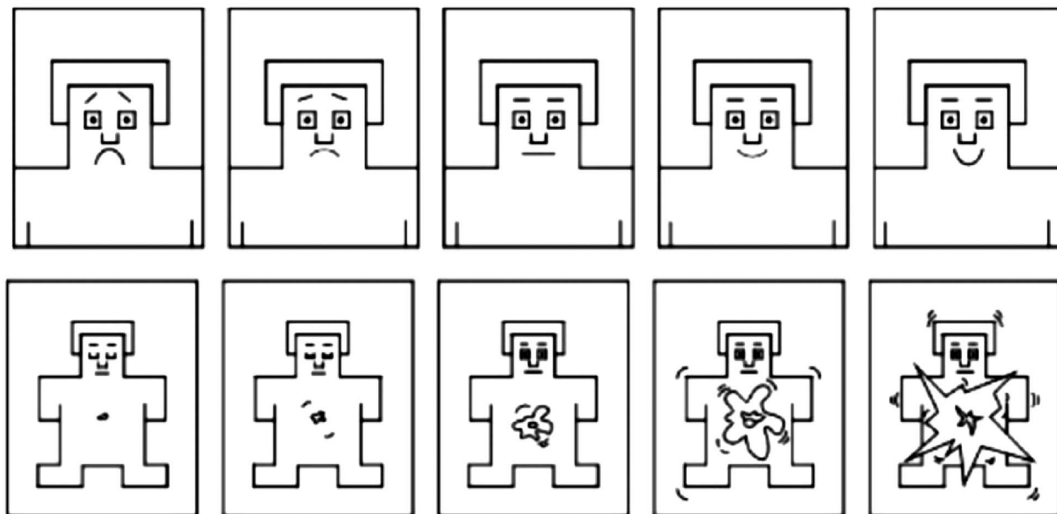
Supplementary Statistical Models and Regression Tables

To evaluate if the association between heart rate synchrony and self-reported measures can be explained in terms of the corresponding values of mean heart rate (HR) and HR variance, we constructed two additional ordinal regression models. These models use one self-

reported subjective measure (either arousal or valence) as dependent variable, and use HR synchrony, mean HR and HR variance as independent variables. Additionally, both models control for age and gender. Results of these models are presented in Appendix [Tables A1](#)

Figure A1

Adapted Version of SELF ASSESSMENT MANIKIN Used in the Study



Note. The adapted version of the SELF ASSESSMENT MANIKIN: From “Measuring emotion: The self-assessment manikin and the semantic differential,” by M. M. Bradley and P. J. Lang, 1994, *Journal of Behavior Therapy and Experimental Psychiatry*, 25(1), p. 51 ([https://doi.org/10.1016/0005-7916\(94\)90063-9](https://doi.org/10.1016/0005-7916(94)90063-9)). Copyright 1994 by Elsevier. Adapted with permission; Lang, 1980, used in the study. In the present study, the scale is reversed (negative-to-positive valence and low-to-high arousal). Each scale is also supplied with 10 circles underneath in which participants could mark their state with an “X.” Each scale is also supplied with labels at each endpoint. The endpoints of the scale for arousal are labeled “extreme calm—extreme arousal” and for valence labeled “extremely unpleasant—extremely pleasant.” All questions and labels are in Danish.

(Appendix continues)

Table A1*Model: HR Synchrony Predicting Self-Reported Arousal (With Covariates)*

Model	Estimate (β)	SE	<i>p</i>
HR synchrony	0.28	0.16	.073
Mean HR	0.69	0.16	<.001
HR variance	0.35	0.16	.031
Age	0.027	0.15	.857
Gender (male)	-1.00	0.26	<.001

Note. Results from an ordinal regression predicting self-reported arousal in terms of HR markers, when also including mean HR and HR variance (for a comparison, see the results of Model 1a). HR = heart rate; SE = standard error.

Table A2*Model: HR Synchrony Predicting Self-Reported Valence (With Covariates)*

Model	Estimate (β)	SE	<i>p</i>
HR synchrony	-0.11	0.18	.524
Mean HR	-0.28	0.15	.064
HR variance	-0.32	0.17	.058
Age	-0.045	0.16	.782
Gender (male)	0.809	0.276	.003

Note. Results from an ordinal regression predicting self-reported valence in terms of HR markers, when also including mean HR and HR variance (for a comparison, see the results of Model 1b). HR = heart rate; SE = standard error.

Table A3*Model: HR Synchrony Predicting Self-Reported Arousal (No Covariates)*

Model	Estimate (β)	SE	<i>p</i>
HR synchrony	0.55	0.12	<.001

Note. Results from an ordinal regression predicting subjective arousal in terms of HR synchrony, without considering controlling variables (but including group effect as random intercept). For a comparison, see the results of Model 1a. HR = heart rate; SE = standard error.

and A2. Crucially, results show that there is still an effect of HR synchrony over self-reported arousal even after including mean HR and HR variance into the model. In contrast, the effect of HR

Table A4*Model: HR Synchrony Predicting Self-Reported Valence (No Covariates)*

Model	Estimate (β)	SE	<i>p</i>
HR synchrony	-0.40	0.15	<.001

Note. Results from an ordinal regression predicting subjective valence in terms of HR synchrony, without considering controlling variables (but including group effect as random intercept). For a comparison, see the results of Model 1b. HR = heart rate; SE = standard error.

Table A5*Model: Social Closeness Predicting HR Synchrony (No Covariates)*

Model	Estimate (β)	SE	<i>p</i>
Close	0.30	0.11	.006

Note. Results from a linear mixed effect model predicting HR synchrony in terms of if people were socially close, without considering controlling variables (but including group effect as random intercept). For a comparison, see results of Model 2a. SE = standard error; HR = heart rate.

Table A6*Model: Familiarity Predicting HR Synchrony (No Covariates)*

Model	Estimate (β)	SE	<i>p</i>
Know	0.30	0.13	.024

Note. Results from a linear mixed effect model predicting HR synchrony in terms of if people know each other, without considering controlling variables (but including group effect as random intercept). For a comparison, see the results of Model 2b. SE = standard error; HR = heart rate.

synchrony over valence goes away. Interestingly, the association between age and both arousal and valence also goes away, suggesting that this can be better explained by the physiological information provided by the HR markers.

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